# Configure role-based access control (RBAC)

SnapCenter Software 4.7

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# Table of Contents

Configure role-based access control (RBAC) . . . 1
Add a user or group and assign role and assets . . . 1
Create a role . . . 3
Add an ONTAP RBAC role using security login commands . . . 4
Create SVM roles with minimum privileges . . . 6
Create ONTAP cluster roles with minimum privileges . . . 10
Configure IIS Application Pools to enable Active Directory read permissions . . . 16

# Configure role-based access control (RBAC)

## Add a user or group and assign role and assets

To configure role-based access control for SnapCenter users, you can add users or groups and assign role. The role determines the options that SnapCenter users can access.

**What you will need**

- You must have logged in as the "SnapCenterAdmin" role.
- You must have created the user or group accounts in Active Directory in the operating system or database. You cannot use SnapCenter to create these accounts.

> From SnapCenter 4.5, you can include only the following special characters in user names and group names: space ( ), hyphen (-), underscore (_), and colon (:). If you want to use a role that you created in an earlier release of SnapCenter with these special characters, you can disable the validation of the role name by changing the value of 'DisableSQLInjectionValidation' parameter to true in the web.config file located where the SnapCenter WebApp is installed. After modifying the value, you do not have to restart the service.

- SnapCenter includes several predefined roles.

  You can either assign these roles to the user or create new roles.

- AD Users and AD Groups that are added to SnapCenter RBAC must have the READ permission on the Users Container and the Computers Container in the Active Directory.
- After you assign a role to a user or group that contains the appropriate permissions, you must assign the user access to SnapCenter assets, such as hosts and storage connections.

  This enables users to perform the actions for which they have permissions on the assets that are assigned to them.

- You should assign a role to the user or group at some point to take advantage of RBAC permissions and efficiencies.
- You can assign assets like host, resource groups, policy, storage connection, plug-in, and credential to the user while creating the user or group.
- The minimum assets that you should assign an user to perform certain operations are as follows:

| Operation | Assets assignment |
|---|---|
| Protect resources | host, policy |
| Backup | host, resource group, policy |
| Restore | host, resource group |

| Operation | Assets assignment |
| --- | --- |
| Clone | host, resource group, policy |
| Clone lifecycle | host |
| Create a Resource Group | host |

- When a new node is added to a Windows cluster or a DAG (Exchange Server Database Availability Group) asset and if this new node is assigned to a user, you must reassign the asset to the user or group to include the new node to the user or group.

  You should reassign the RBAC user or group to the cluster or DAG to include the new node to the RBAC user or group. For example, you have a two-node cluster and you have assigned an RBAC user or group to the cluster. When you add another node to the cluster, you should reassign the RBAC user or group to the cluster to include the new node for the RBAC user or group.

- If you are planning to replicate Snapshot copies, you must assign the storage connection for both the source and destination volume to the user performing the operation.

  You should add assets before assigning access to the users.

> If you are using the SnapCenter Plug-in for VMware vSphere functions to protect VMs, VMDKs, or datastores, you should use the VMware vSphere GUI to add a vCenter user to a SnapCenter Plug-in for VMware vSphere role. For information about VMware vSphere roles, see Predefined roles packaged with SnapCenter Plug-in for VMware vSphere.

**Steps**

1. In the left navigation pane, click **Settings**.
2. In the Settings page, click **Users and Access** > +.
3. In the Add Users/Groups from Active Directory or Workgroup page:

| For this field… | Do this… |
| --- | --- |
| Access Type | Select either Domain or workgroup<br><br>For Domain authentication type, you should specify the domain name of the user or group to which you want to add the user to a role.<br><br>By default, it is pre-populated with the logged in domain name.<br><br>You must register the untrusted domain in the **Settings** > **Global Settings** > **Domain Settings** page. |

| For this field… | Do this… |
| --- | --- |
| Type | Select either User or Group<br><br>SnapCenter supports only security group and not the distribution group. |
| User Name | a. Type the partial user name, and then click **Add**.<br><br>The user name is case-sensitive.<br><br>b. Select the user name from the search list.<br><br>When you add users from a different domain or an untrusted domain, you should type the user name fully because there is no search list for cross domain users.<br><br>Repeat this step to add additional users or groups to the selected role. |
| Roles | Select the role to which you want to add the user. |

4. Click **Assign**, and then in the Assign Assets page:
   a. Select the type of asset from the **Asset** drop-down list.
   b. In the Asset table, select the asset.

      The assets are listed only if the user has added the assets to SnapCenter.

   c. Repeat this procedure for all of the required assets.
   d. Click **Save**.
5. Click **Submit**.

After adding users or groups and assigning roles, refresh the resources list.

# Create a role

In addition to using the existing SnapCenter roles, you can create your own roles and customize the permissions.

You should have logged in as the "SnapCenterAdmin" role.

**Steps**

1. In the left navigation pane, click **Settings**.
2. In the Settings page, click **Roles**.

3. Click +.
4. In the Add Role page, specify a name and description for the new role.

> From SnapCenter 4.5, you can include only the following special characters in user names and group names: space ( ), hyphen (-), underscore (_), and colon (:). If you want to use a role that you created in an earlier release of SnapCenter with these special characters, you can disable the validation of the role name by changing the value of 'DisableSQLInjectionValidation' parameter to true in the web.config file located where the SnapCenter WebApp is installed. After modifying the value, you do not have to restart the service.

5. Select **All members of this role can see other members' objects** to enable other members of the role to see resources such as volumes and hosts after they refresh the resources list.

   You should deselect this option if you do not want members of this role to see objects to which other members are assigned.

   > When this option is enabled, assigning users access to objects or resources is not required if users belong to the same role as the user who created the objects or resources.

6. In the Permissions page, select the permissions that you want to assign to the role, or click **Select All** to grant all permissions to the role.
7. Click **Submit**.

# Add an ONTAP RBAC role using security login commands

You can use the security login commands to add an ONTAP RBAC role when your storage systems are running clustered ONTAP.

### What you will need

- Before you create an ONTAP RBAC role for storage systems running clustered ONTAP, you must identify the following:
  - The task (or tasks) that you want to perform
  - The privileges required to perform these tasks
- Configuring an RBAC role requires that you perform the following actions:
  - Grant privileges to commands and/or command directories.

    There are two levels of access for each command/command directory: all-access and read-only.

    You must always assign the all-access privileges first.

  - Assign roles to users.
  - Vary your configuration depending on whether your SnapCenter plug-ins are connected to the Cluster Administrator IP for the entire cluster or directly connected to a SVM within the cluster.

### About this task

To simplify configuring these roles on storage systems, you can use the RBAC User Creator for Data ONTAP tool, which is posted on the NetApp Communities Forum.

This tool automatically handles setting up the ONTAP privileges correctly. For example, RBAC User Creator for Data ONTAP tool automatically adds the privileges in the correct order so that the all-access privileges appear first. If you add the read-only privileges first and then add the all-access privileges, ONTAP marks the all-access privileges as duplicates and ignores them.

> If you later upgrade SnapCenter or ONTAP, you should re-run the RBAC User Creator for Data ONTAP tool to update the user roles you created previously. User roles created for an earlier version of SnapCenter or ONTAP do not work properly with upgraded versions. When you re-run the tool, it automatically handles the upgrade. You do not need to recreate the roles.

More information about setting up ONTAP RBAC roles, see the ONTAP 9 Administrator Authentication and RBAC Power Guide.

> For consistency, the SnapCenter documentation refers to the roles as using privileges. The OnCommand System Manager GUI uses the term *attribute* instead of *privilege*. When setting up ONTAP RBAC roles, both these terms mean the same thing.

**Steps**

1. On the storage system, create a new role by entering the following command:

   ```
   security login role create <role_name\> -cmddirname "command" -access all
   -vserver <svm_name\>
   ```

   - svm_name is the name of the SVM. If you leave this blank, it defaults to cluster administrator.
   - role_name is the name you specify for the role.
   - command is the ONTAP capability.

     > You must repeat this command for each permission. Remember that all-access commands must be listed before read-only commands.

     For information about the list of permissions, see ONTAP CLI commands for creating roles and assigning permissions.

2. Create a user name by entering the following command:

   ```
   security login create -username <user_name\> -application ontapi -authmethod
   <password\> -role <name_of_role_in_step_1\> -vserver <svm_name\> -comment
   "user_description"
   ```

   - user_name is the name of the user you are creating.
   - <password> is your password. If you do not specify a password, the system will prompt you for one.
   - svm_name is the name of the SVM.

3. Assign the role to the user by entering the following command:

   ```
   security login modify username <user_name\> -vserver <svm_name\> -role
   <role_name\> -application ontapi -application console -authmethod <password\>
   ```

   - <user_name> is the name of the user you created in Step 2. This command lets you modify the user to associate it with the role.

- <svm_name> is the name of the SVM.
- <role_name> is the name of the role you created in Step 1.
- <password> is your password. If you do not specify a password, the system will prompt you for one.

4. Verify that the user was created correctly by entering the following command:

```
security login show –vserver <svm_name\> -user-or-group-name <user_name\>
```

user_name is the name of the user you created in Step 3.

# Create SVM roles with minimum privileges

There are several ONTAP CLI commands you must run when you create a role for a new SVM user in ONTAP. This role is required if you configure SVMs in ONTAP to use with SnapCenter and you do not want to use the vsadmin role.

**Steps**

1. On the storage system, create a role and assign all the permissions to the role.

```
security login role create –vserver <svm_name\>- role <SVM_Role_Name\>
-cmddirname <permission\>
```

> You should repeat this command for each permission.

2. Create a user and assign the role to that user.

```
security login create -user <user_name\> -vserver <svm_name\> -application
ontapi -authmethod password -role <SVM_Role_Name\>
```

3. Unlock the user.

```
security login unlock -user <user_name\> -vserver <svm_name\>
```

## ONTAP CLI commands for creating SVM roles and assigning permissions

There are several ONTAP CLI commands you should run to create SVM roles and assign permissions.

- `security login role create -role SVM_Role_Name -cmddirname "snapmirror list-destinations" -vserver SVM_Name -access all`
- `security login role create -role SVM_Role_Name -cmddirname "event generate-autosupport-log" -vserver SVM_Name -access all`
- `security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "job history show" -access all`
- `security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "job stop" -access all`
- `security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "lun" -access all`

- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun create" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun delete" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun igroup add" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun igroup create" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun igroup delete" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun igroup rename" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun igroup show" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun mapping add-reporting-nodes" -access all`
- `security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "lun mapping create" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun mapping delete" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun mapping remove-reporting-nodes" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun mapping show" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun modify" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun move-in-volume" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun offline" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun online" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun resize" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun serial" -access all`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "lun show" -access all`
- `security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "network interface" -access readonly`
- `security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname`

"snapmirror policy add-rule" -access all

- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror policy modify-rule" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror policy remove-rule" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror policy show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror restore" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror show" -access all
- security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "snapmirror show-history" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror update" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "snapmirror update-ls-set" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "version" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume clone create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume clone show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume clone split start" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume clone split stop" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume destroy" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume file clone create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume file show-disk-usage" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume modify" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume offline" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume online" -access all

- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume qtree create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume qtree delete" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume qtree modify" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume qtree show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume restrict" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot delete" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot modify" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot rename" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot restore" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot restore-file" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume snapshot show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "volume unmount" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver cifs share create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver cifs share delete" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver cifs share show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver cifs show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver export-policy create" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver export-policy delete" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname

"vserver export-policy rule create" -access all

- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver export-policy rule show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver export-policy show" -access all
- security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "vserver iscsi connection show" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver" -access readonly
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver export-policy" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname "vserver iscsi" -access all
- security login role create -vserver SVM_Name -role SVM_Role_Name -cmddirname "volume clone split status" -access all
- security login role create -vserver SVM_name -role SVM_Role_Name -cmddirname “volume managed-feature” -access all

# Create ONTAP cluster roles with minimum privileges

You should create an ONTAP cluster role with minimum privileges so that you do not have to use the ONTAP admin role to perform operations in SnapCenter. You can run several ONTAP CLI commands to create the ONTAP cluster role and assign minimum privileges.

**Steps**

1. On the storage system, create a role and assign all the permissions to the role.

   ```
   security login role create –vserver <cluster_name\>- role <role_name\>
   -cmddirname <permission\>
   ```

   You should repeat this command for each permission.

2. Create a user and assign the role to that user.

   ```
   security login create -user <user_name\> -vserver <cluster_name\> -application
   ontapi -authmethod password -role <role_name\>
   ```

3. Unlock the user.

   ```
   security login unlock -user <user_name\> -vserver <cluster_name\>
   ```

## ONTAP CLI commands for creating cluster roles and assigning permissions

There are several ONTAP CLI commands you should run to create cluster roles and assign permissions.

- security login role create -vserver Cluster_name or cluster_name -role Role_Name -cmddirname "metrocluster show" -access readonly
- security login role create -vserver Cluster_name or cluster_name -role Role_Name -cmddirname "cluster identity modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "cluster identity show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "cluster modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "cluster peer show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "cluster show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "event generate-autosupport-log" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "job history show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "job stop" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup add" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup rename" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun igroup show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun mapping add-reporting-nodes" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun mapping create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname

"lun mapping delete" -access all

- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun mapping remove-reporting-nodes" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun mapping show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun move-in-volume" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun offline" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun online" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun persistent-reservation clear" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun resize" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun serial" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "lun show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "network interface create" -access readonly
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "network interface delete" -access readonly
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "network interface modify" -access readonly
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "network interface show" -access readonly
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "security login" -access readonly
- security login role create -role Role_Name -cmddirname "snapmirror create" -vserver Cluster_name -access all
- security login role create -role Role_Name -cmddirname "snapmirror list-destinations" -vserver Cluster_name -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy add-rule" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy delete" -access all

- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy modify-rule" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy remove-rule" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror policy show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror restore" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror show-history" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror update" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "snapmirror update-ls-set" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system license add" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system license clean-up" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system license delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system license show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system license status show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system node modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system node show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "system status show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "version" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume clone create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume clone show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname

"volume clone split start" -access all

- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume clone split stop" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume destroy" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume file clone create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume file show-disk-usage" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume offline" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume online" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume qtree create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume qtree delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume qtree modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume qtree show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume restrict" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot promote" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot rename" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot restore" -access all

- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot restore-file" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume snapshot show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "volume unmount" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs share modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs share create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs share delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs share modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs share show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver cifs show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy rule create" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy rule delete" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy rule modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver export-policy rule show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname

"vserver export-policy show" -access all

- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver iscsi connection show" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver modify" -access all
- security login role create -vserver Cluster_name -role Role_Name -cmddirname "vserver show" -access all

# Configure IIS Application Pools to enable Active Directory read permissions

You can configure Internet Information Services (IIS) on your Windows Server to create a custom Application Pool account when you need to enable Active Directory read permissions for SnapCenter.

**Steps**

1. Open IIS Manager on the Windows Server where SnapCenter is installed.
2. In the left navigation pane, click **Application Pools**.
3. Select SnapCenter in the Application Pools list, and then click **Advanced Settings** in the Actions pane.
4. Select Identity, and then click **…** to edit the SnapCenter application pool identity.
5. In the Custom Account field, enter a domain user or domain admin account name with Active Directory read permission.
6. Click OK.

   The custom account replaces the built-in ApplicationPoolIdentity account for the SnapCenter application pool.

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